



大连理工大学
DALIAN UNIVERSITY OF TECHNOLOGY

School of Mathematical Sciences
Dalian University of Technology

Linggong Road 2
116024 Dalian, P.R. China
dtdynamics.github.io/ddes



DDES - DUT Differential Equations Seminar
Seminar Room 114
9:00am - 9:45am, Thursday, October 8, 2026

The linear stability of relative equilibria in the N-body problem

Jiexin Sun (孙洁昕, PostDoc@DUT)

Abstract. In this talk, I will introduce some results on the linear stability of relative equilibria in the N-body problem obtained using the Maslov-type index. The talk consists of three parts. In the first part, I will briefly introduce the basic definition of a relative equilibrium and describe the linear stability problem. In the second part, I will introduce some basic background on the Maslov-type index and explain the idea of using the Maslov-type index to study the linear stability problem. In the third part, I will present the fundamental solution of the Kepler problem and then some qualitative and quantitative results on the linear stability of relative equilibria in the N-body problem obtained using the idea introduced in the second part. This talk is based on joint work with Prof. Xijun Hu and Prof. Yuwei Ou.

Jiexin Sun

2026.09–Present PostDoc in Mathematics, Dalian University of Technology, China.
2026.03–2026.06 Visiting Scholar, Nankai University, China.
2021.09–2025.12 PhD in Mathematics, Shandong University, China.
2023.11–2025.10 Visiting PhD student, Paris Observatory, France.
2018.09–2021.06 Master in Mathematics, Shandong University, China.
2014.09–2018.06 Bachelor in Mathematics, Lanzhou University, China.

Research Interests: Relative equilibria of the Newtonian N-body problem, Applications of Maslov index theory in the N-body problem.

